

Secondary Student Check-In

Winter 2022

Student Wellbeing

Methods

Animals

All experimental procedures were approved by the University of Florida Institutional Animal Care and Use Committee. The study was conducted and data is reported in accordance with the ARRIVE guidelines. Sprague-Dawley rats [120 males and 72 females (generation F0)] were purchased from Charles River (Wilmington, MA). The F0 female rats were used as breeders only to generate offspring (generation F1). All rats were housed under controlled illumination (12-h light/12-h dark, lights on at 7:00 AM) and temperature (23–24 °C) with free access to food and water. Experimental data in this study are from 192 male and 72 female rats. One rat in injury group died immediately after the traumatic brain injury induction and two other rats in the surgery group were removed from the study because of the suture failure during the recovery period. All three rats were excluded from all analyses.

Treatment Groups

Figure 1 shows an overview of the study design. F0 male rats were randomized into four treatment groups (n = 30/group) using a randomization plan with a web-based generator, and the investigators were blinded to group assignments. Rats in the injury group were subjected to all interventions: 1) surgery under 3% SEVO anesthesia for 40 min on postnatal day 60 to conduct a craniectomy and implant an injury hub; 2) a midline fluid percussion injury inflicted moderate traumatic brain injury^{1,2} on the same day; and 3) exposure to 2.1% sevoflurane for 3 h on postnatal days 62, 64, and 66,³ to model anesthesia/sedation needed for treatment of conditions associated with traumatic brain injury or unrelated

ones. Rats in the sevoflurane group had only sevoflurane exposure on postnatal days 60, 62, 64, and 66. Rats in the surgery (sham) group had a craniectomy and injury hub implantation, but not traumatic brain injury, on postnatal day 60. These rats were exposed to 2.1% sevoflurane for 3 h on postnatal days 62, 64, and 66. Rats in the control group were placed in a new cage and housed one per cage for an equivalent time on postnatal days 60, 62, 64, and 66.

A subset of F0 male rats from all groups ($n = 12/\text{group}$) was sacrificed 1 h after recovery from sevoflurane anesthesia on postnatal day 66 or at an equivalent timepoint in the control group to study acute effects. The remaining F0 male rats ($n = 18/\text{group}$) were mated on postnatal day 90 with F0 control female rats to produce offspring (generation F1). A cohort of 18 breeding pairs of F0 male and female rats were used to produce F1 offspring for a given experimental group (18 F1 rats/group/sex). A given F1 offspring experimental group included 1 to 2 rats from a given F1 litter. Within 24 h of delivery, F1 litters were culled to 12 pups. The F0 females were housed individually throughout the entire gestation and postpartum rearing periods. At the age of 21 days, pups were weaned and housed in sex-matched pairs for the rest of the study. F1 rats were not exposed to any treatment and were subjected to animal facility rearing only.

The F0 sires were sequentially evaluated in the elevated plus maze⁴ on postnatal day 105, for prepulse inhibition of the acoustic startle response^{4,5} on postnatal day 115, and in the Morris water maze^{6,7} between postnatal days 125 and 130. Ten days later, stress-induced release of corticosterone was studied after exposing the animals to physical restraint for 30 min. The serum levels of corticosterone were measured before physical restraint and 10, 60, and 120 min after the restraint. Blood sampling ($\sim 300 \mu\text{L}$) was done using the “tail clip” method, as previously described.³ The F1 male and female offspring were evaluated in the elevated plus maze on postnatal day 60, for prepulse inhibition of the acoustic startle response on postnatal day 70, and in Morris water maze tests between postnatal days 90 and 95. Blood samples were collected using the “tail

clip” method 30 min after completion of the prepulse inhibition test to measure stress-induced serum corticosterone levels. Ten days after completing the *in vivo* studies, the F0 and F1 rats were anesthetized and euthanized through decapitation to collect trunk blood and brain tissue samples for enzyme-linked immunosorbent assay,³ reverse transcription-polymerase chain reaction,^{3,6} immunohistochemistry,³ and targeted next-generation bisulfite sequencing studies.⁸

Midline Fluid Percussion Injury Model of Traumatic Brain Injury

The fluid percussion injury device (Model 01-B, Custom Design & Fabrication, Inc., Sandston, VA, USA) was used to induce the midline fluid percussion injury model of moderate traumatic brain injury.^{1,2} Anesthesia was induced with 6% sevoflurane for 3 min in an anesthesia chamber and then a rat was placed in a stereotaxic frame with a continuous supply of 3% sevoflurane for 37 min via a nose cone. Sevoflurane was delivered in a mixture of ~50% O₂ in air (1.5 L/min). To apply a uniform sevoflurane anesthesia duration and depth to all experimental animals, anesthesia in all rats was induced with 6% sevoflurane for 3 min and maintained with 3% sevoflurane for 37 min, which was slightly longer than the time needed for the craniectomy and injury hub implantation procedure. The rats’ rectal temperature during surgery was monitored and maintained at ~+37.0 °C using a homeothermic monitoring system (50-7220F, Harvard Apparatus, St. Laurent, Quebec, Canada). After a ~15-mm midline incision and fascia removal from the skull, a craniectomy was centrally made over the sagittal suture between bregma and lambda by using a trephine with an outer diameter of 4.8 mm (Integra Miltex, Princeton, NJ, USA). The dura was kept intact. The female hub of a 20-